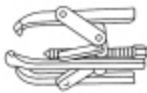
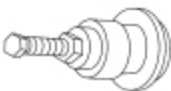


Crankshaft Front Seal

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST1184-A	3-Jaw Puller or equivalent 303-D121
 ST2132-A	Adapter Set, Step Plate 205-DS011 or equivalent
 ST3245-A	Installer, Crankshaft Damper and Crankshaft Front Seal 303-1508
 ST1328-A	Installer, Front Cover Oil Seal 303-335 (T88T-6701-A)
 ST1385-A	Remover, Oil Seal 303-409 (T92C-6700-CH)
 ST1438-A	Strap Wrench 303-D055 (D85L-6000-A) or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

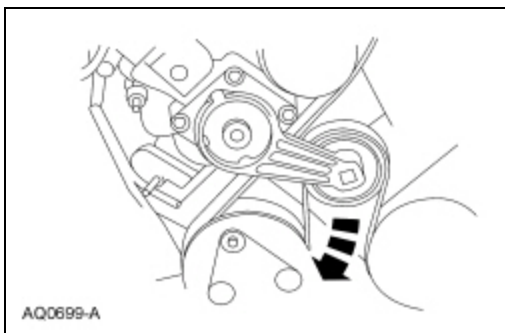
Material

Item	Specification

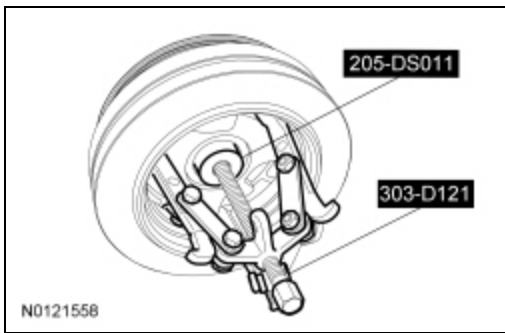
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS- M2C945-A
Motorcraft® Silicone Gasket Remover ZC-30-A	—
Motorcraft® Silicone Gasket and Sealant TA-30	WSE- M4G323-A4

Removal

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. Rotate the tensioner clockwise and remove the accessory drive belt from the crankshaft pulley.

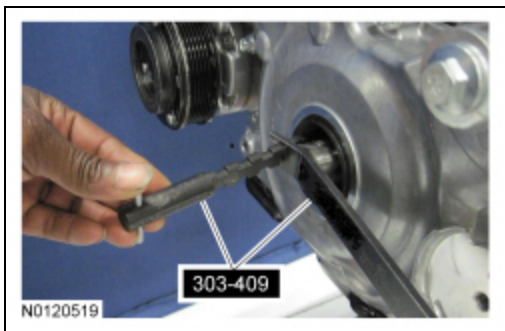


3. Remove and discard the crankshaft pulley bolt.
4. Using the 3 Jaw Puller and Step Plate, remove the crankshaft pulley.



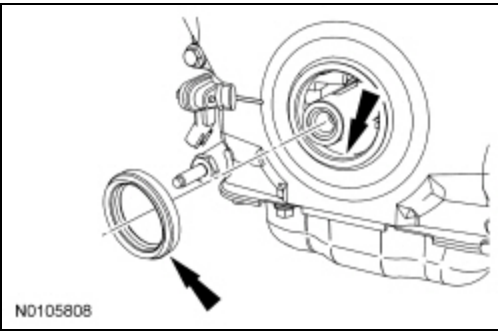
5. **NOTICE:** Use care not to damage the engine front cover or the crankshaft when removing the seal.

Using the Oil Seal Remover, remove the crankshaft front oil seal.

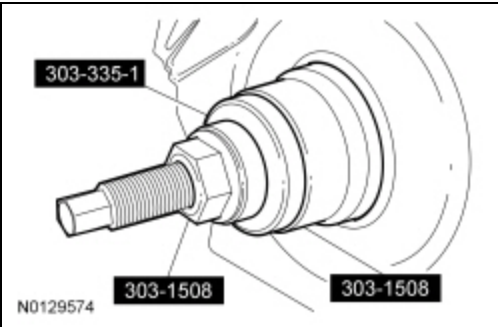


Installation

1. Lubricate the engine front cover and the new crankshaft seal inner lip with clean engine oil.

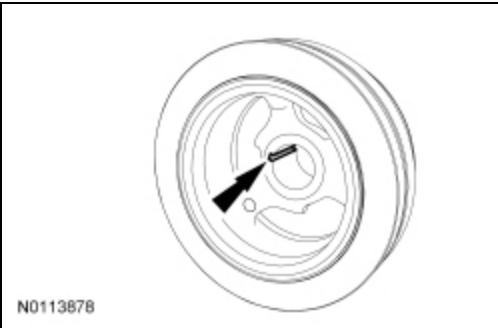


2. Using the Crankshaft Damper and Crankshaft Front Seal Installer and the Front Cover Oil Seal Installer, install the new crankshaft front seal into the engine front cover.

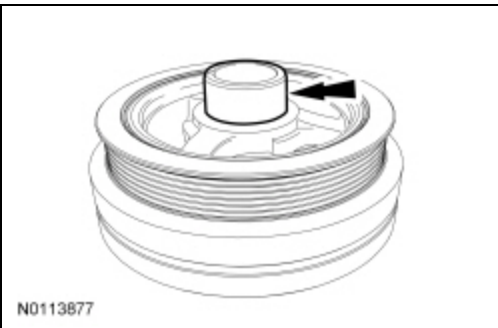


3. **NOTE:** *If not secured within 5 minutes, the sealant must be removed and the sealing area cleaned with metal surface prep and silicone gasket remover. Allow to dry until there is no sign of wetness, or 5 minutes, whichever is longer. Failure to follow this procedure can cause future oil leakage.*

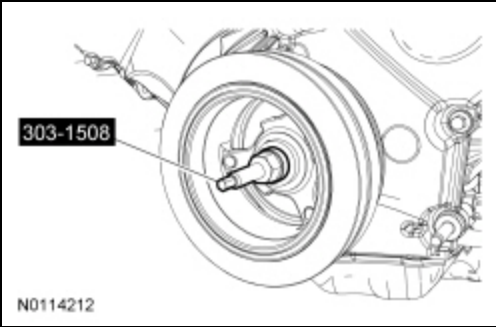
Apply silicone gasket and sealant to the Woodruff key slot in the crankshaft pulley.



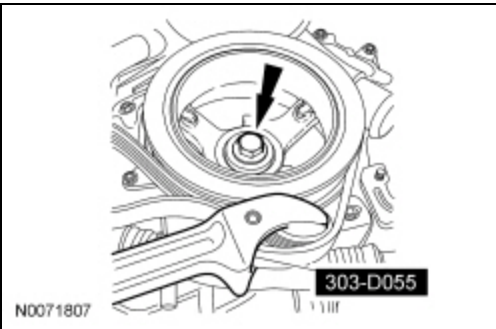
4. Lubricate the crankshaft pulley sealing area with clean engine oil prior to installation.



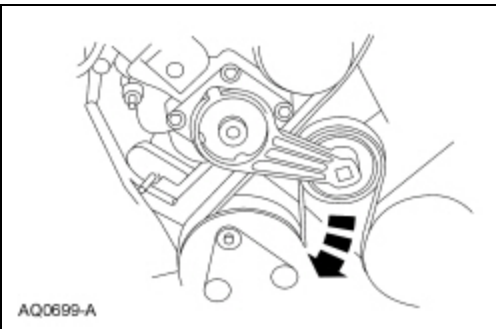
5. Using the Crankshaft Vibration Damper Installer, install the crankshaft pulley.



6. Using the Strap Wrench, install a new crankshaft pulley bolt and the original washer, tighten the bolt in 2 stages:
- Stage 1: Tighten to 175 Nm (129 lb-ft).
 - Stage 2: Tighten an additional 90 degrees.



7. Rotate the tensioner clockwise and install the accessory drive belt onto the crankshaft pulley.



8. After completing the repairs, use the scan tool to perform the Misfire Monitor Neutral Profile Correction procedure following the on-screen instructions.